3.4 Mechanics and Materials

1. Scalars and Vectors

Scalar: A quantity with magnitude only. Examples: mass, temperature, speed, energy, distance, time, power, pressure.
Vector: A quantity with both magnitude and direction. Examples: force, velocity, displacement, acceleration, momentum, weight, electric field strength.

Resolving Vectors

Any vector can be resolved into two perpendicular components. If the vector F makes angle θ with the horizontal:

- Horizontal component: $F_x = F \cos\theta$
- Vertical component: $F_y = F \sin\theta$

The magnitude of the resultant: $R = \sqrt{(F_x^2 + F_y^2)}$. The angle: $\theta = \tan^{-1}(F_y/F_x)$.

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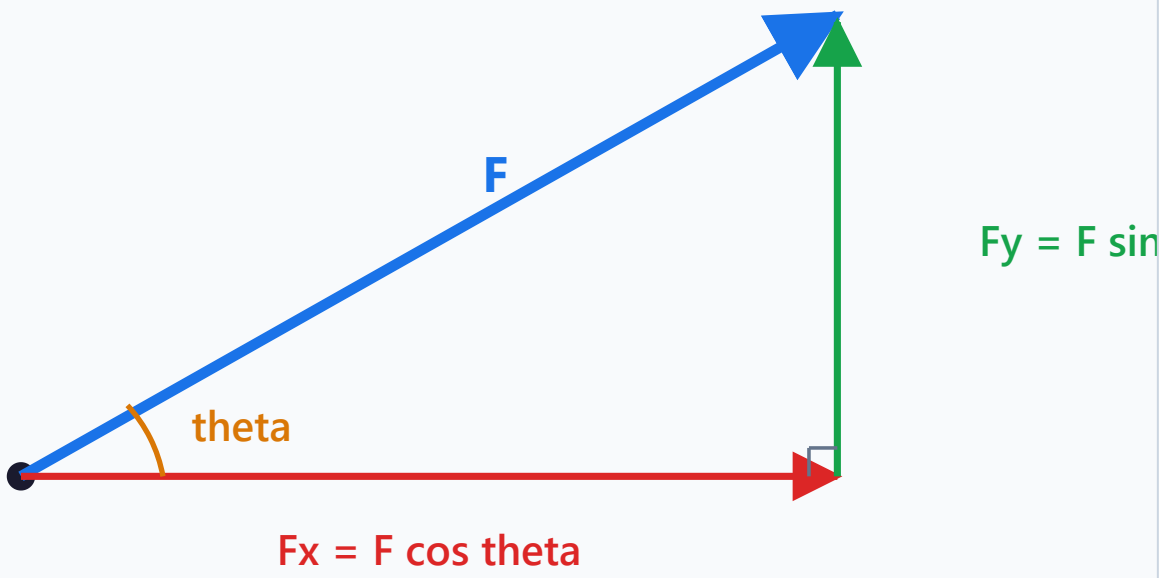
Resolving a Force into Components



R = sqrt(Fx2 + Fy2), theta = tan-1(Fy / Fx)

Worked Example: A force of 50 N acts at 35° above the horizontal. Find its horizontal and vertical components.

F_x = 50 cos 35° = 50 × 0.8192 = **41.0 N**
F_y = 50 sin 35° = 50 × 0.5736 = **28.7 N**

Check: $\sqrt{(41.0^2 + 28.7^2)} = \sqrt{(1681 + 823)} = \sqrt{(2504)} = 50.0$ N ✓

Adding Vectors

Vectors are added tip-to-tail. For calculation: resolve each vector into components, add the x-components and y-components separately, then recombine.

Worked Example: Two forces act on an object: 30 N east and 40 N north. Find the resultant.

$R_x = 30$ N, $R_y = 40$ N

$R = \sqrt{(30^2 + 40^2)} = \sqrt{(900 + 1600)} = \sqrt{2500} =$ **50 N**

$\theta = \tan^{-1}(40/30) = \tan^{-1}(1.333) =$ **53.1° north of east**

Exam Tip: When resolving forces on an inclined plane, resolve parallel and perpendicular to the slope (not horizontal and vertical). The component of weight parallel to the slope is mg sinθ, and perpendicular is mg cosθ, where θ is the angle of the slope.

2. Moments

Moment: The turning effect of a force about a pivot. Moment = F × d, where d is the perpendicular distance from the line of action of the force to the pivot. Units: N m.

Principle of Moments

For a body in rotational equilibrium, the sum of clockwise moments about any point equals the sum of anticlockwise moments about that same point.

Worked Example: A uniform beam of length 4.0 m and weight 200 N is supported at both ends. A 500 N load is placed 1.0 m from end A. Find the reaction forces at A and B.

Taking moments about A (eliminating R_A):

Clockwise moments = Anticlockwise moments about A:

$500 \times 1.0 + 200 \times 2.0 = R_B \times 4.0$

$500 + 400 = 4.0\ R_B$

$R_B = 900 / 4.0 =$ **225 N**

Resolving vertically: $R_A + R_B = 500 + 200$

$R_A = 700 - 225 =$ **475 N**

Couples

A couple is a pair of equal and opposite forces whose lines of action do not coincide. A couple produces a pure turning effect (torque) with no resultant linear force.

$$\text{Torque of a couple} = F \times d$$

where d is the perpendicular distance between the lines of action of the forces. A couple causes rotation without translation.

Real-world application: Turning a steering wheel, opening a tap, and using a screwdriver all involve couples. In engineering, torque wrenches are used to apply a precise couple to bolts, which is critical in aircraft assembly and engine construction.

Equilibrium

For a body to be in static equilibrium:

- The resultant force in any direction is zero (translational equilibrium): $\Sigma F = 0$
- The resultant moment about any point is zero (rotational equilibrium): $\Sigma M = 0$

3. Kinematics (SUVAT)

Displacement (s): Vector distance from start to finish in a specified direction. Units: m.
Velocity (v): Rate of change of displacement. $v = ds/dt$. Units: $m\ s^{-1}$.
Acceleration (a): Rate of change of velocity. $a = dv/dt$. Units: $m\ s^{-2}$.

Graphs of Motion

Graph	Gradient gives	Area under curve gives
Displacement-time	Velocity	—
Velocity-time	Acceleration	Displacement
Acceleration-time	—	Change in velocity

SUVAT Equations (constant acceleration only)

Equation	Missing Variable

$v = u + at$	s
$s = ut + \frac{1}{2}at^2$	v
$s = vt - \frac{1}{2}at^2$	u
$v^2 = u^2 + 2as$	t
$s = \frac{1}{2}(u + v)t$	a

Derivation of $v^2 = u^2 + 2as$: From $v = u + at$, we get $t = (v - u)/a$. Substituting into $s = \frac{1}{2}(u + v)t$: $s = \frac{1}{2}(u + v)(v - u)/a = (v^2 - u^2)/(2a)$. Rearranging: $v^2 = u^2 + 2as$.

Worked Example: A car travelling at 30 m s^{-1} brakes with a deceleration of 5.0 m s^{-2}. Find (a) the stopping time, (b) the stopping distance.

$u = 30$ m s^{-1}, $v = 0$, $a = -5.0$ m s^{-2}

(a) $v = u + at$: $0 = 30 + (-5.0)t$, so $t = 30/5.0 =$ **6.0 s**

(b) $v^2 = u^2 + 2as$: $0 = 30^2 + 2(-5.0)s$
$0 = 900 - 10s$, so $s = 900/10 =$ **90 m**

Free Fall

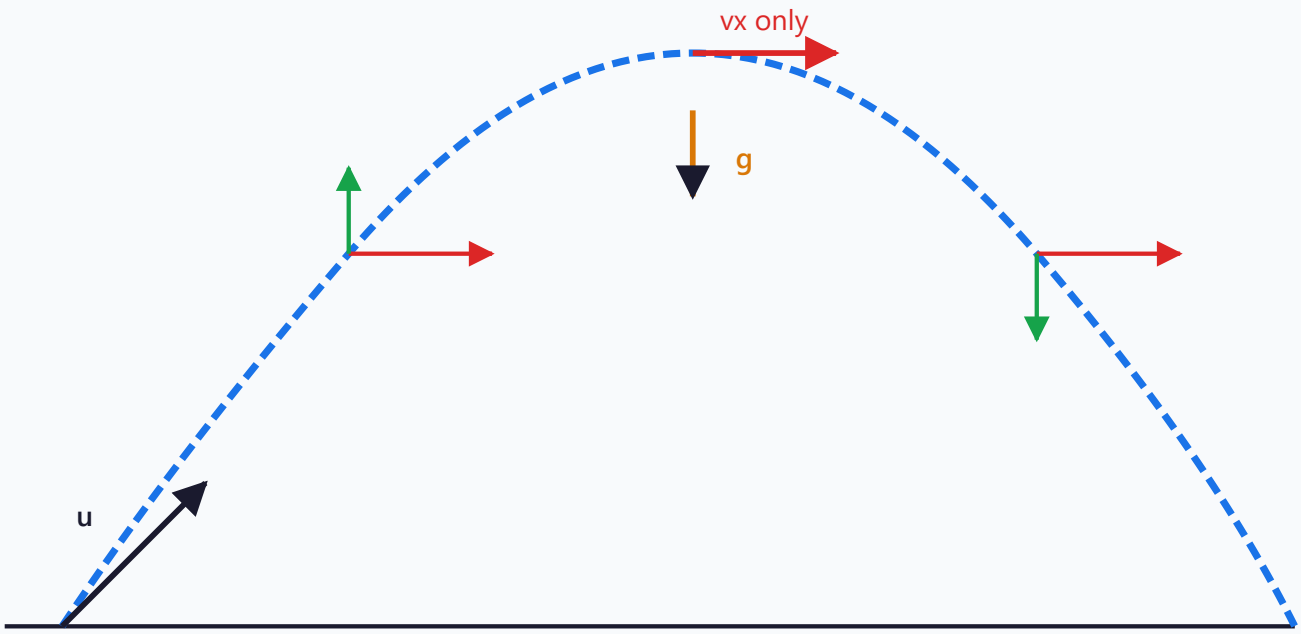
An object in free fall near the Earth's surface has acceleration $a = g = 9.81$ m s^{-2} (downward). Air resistance is neglected. All objects fall with the same acceleration regardless of mass (Galileo's insight, confirmed on the Moon where there is no air resistance).

Projectile Motion

A projectile has two independent components of motion:

- **Horizontal:** Constant velocity (no horizontal acceleration, ignoring air resistance). $s_x = v_x t$.
- **Vertical:** Constant acceleration due to gravity ($a = g = 9.81$ m s^{-2} downward). Use SUVAT.
- **Time** is the link between the two components — both share the same elapsed time.

Projectile Motion



Horizontal velocity (vx) is constant. Vertical velocity (vy) changes due to gravity.

Worked Example: A ball is launched horizontally at 15 m s^{-1} from the top of a 20 m high cliff. Find (a) the time to reach the ground, (b) the horizontal distance travelled, (c) the speed on impact.

(a) Vertical: $s = ut + ½gt^2$. $u_y = 0$ (launched horizontally), s = 20 m, g = 9.81 m s^{-2}

$20 = 0 + ½(9.81)t^2$

$t^2 = 40/9.81 = 4.077$

t = **2.02 s**

(b) Horizontal: $x = v_x t = 15 \times 2.02 =$ **30.3 m**

(c) $v_y = u_y + gt = 0 + 9.81 \times 2.02 = 19.8$ m s^{-1}

$v_x = 15$ m s^{-1} (unchanged)

Speed = $\sqrt{(15^2 + 19.8^2)} = \sqrt{(225 + 392)} = \sqrt{617} =$ **24.8 m s^{-1}**

Angle below horizontal = $\tan^{-1}(19.8/15) = 52.8°$

Exam Tip: For projectiles, always resolve the initial velocity into horizontal (u cosθ) and vertical (u sinθ) components, then treat each direction separately. Time is the link between the two. At the peak, the vertical velocity is zero but the horizontal velocity is still u cosθ. The speed at the peak is NOT zero.

4. Newton's Laws of Motion

Newton's First Law: An object remains at rest or continues to move at constant velocity unless acted on by a resultant external force. This defines inertia — the tendency of an object to resist changes in its motion.

Newton's Second Law: The rate of change of momentum of an object is directly proportional to the resultant force acting on it and takes place in the direction of that force. For constant mass: F = ma.

Newton's Third Law: If object A exerts a force on object B, then object B exerts an equal and opposite force on object A. The two forces: (1) are equal in magnitude, (2) opposite in direction, (3) the same type of force, (4) act on different objects, (5) exist simultaneously.

Worked Example: A 1200 kg car accelerates from rest to 25 m s^{-1} in 8.0 s. Calculate the resultant force, assuming it is constant.

a = (v - u)/t = (25 - 0)/8.0 = 3.125 m s^{-2}
F = ma = 1200 × 3.125 = **3750 N** (= 3.8 kN to 2 s.f.)

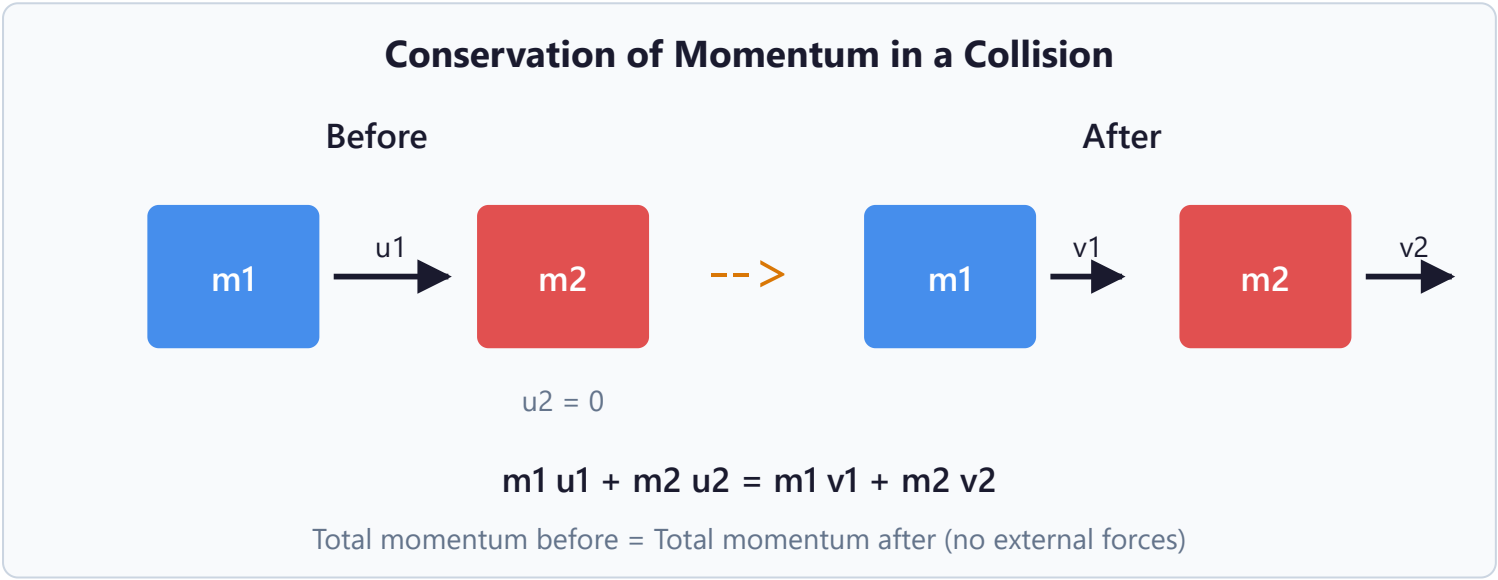
Exam Tip: Newton's third law pairs must act on DIFFERENT objects and be the SAME type of force. Weight (Earth pulls object) and normal contact force (table pushes object) are NOT a third law pair because both act on the same object. The third law pair to the weight is the gravitational pull of the object on the Earth.

5. Momentum and Impulse

$$p = mv$$

Momentum is a vector quantity. Units: kg m s^{-1} or N s.

Conservation of linear momentum: In a closed system where no external resultant force acts, the total momentum before an event equals the total momentum after the event. This applies to all collisions and explosions.



Impulse

$$\text{Impulse} = F\Delta t = \Delta p = mv - mu$$

Impulse equals the change in momentum. On a force-time graph, impulse is the area under the curve. Units: N s = kg m s^{-1}.

Worked Example: A 0.45 kg football moving at 12 m s^{-1} is kicked and its velocity changes to 20 m s^{-1} in the same direction. The foot is in contact with the ball for 0.05 s. Find the average force exerted.

Impulse = Δp = m(v - u) = 0.45 × (20 - 12) = 0.45 × 8 = 3.6 N s
F = Δp / Δt = 3.6 / 0.05 = **72 N**

Real-world application: Car crumple zones and airbags increase the time of collision, reducing the force on the occupants (same impulse but over a longer time). Similarly, catching a cricket ball by moving your hands backward extends the stopping time, reducing the impact force.

Types of Collision

Type	Momentum Conserved?	Kinetic Energy Conserved?	Example
Elastic	Yes	Yes	Atomic/molecular collisions, Newton's cradle (approximately)
Inelastic	Yes	No (KE lost to heat, sound, deformation)	Most real-world collisions
Perfectly inelastic	Yes	No (objects stick together, maximum KE loss)	Bullet embedding in a block
Explosion	Yes (total = 0 if from rest)	No (KE increases, from internal energy)	Cannon recoil, fireworks

Worked Example: A 10 g bullet travelling at 400 m s^{-1} embeds in a stationary 2.0 kg wooden block. Find the velocity of the block+bullet after impact and the kinetic energy lost.

Conservation of momentum: $m_1u_1 = (m_1 + m_2)v$

$0.010 \times 400 = (0.010 + 2.0) \times v$

$4.0 = 2.01 \times v$

$v = 4.0 / 2.01 =$ **1.99 m s^{-1}**

KE before = $\frac{1}{2}(0.010)(400^2) = 800$ J

KE after = $\frac{1}{2}(2.01)(1.99^2) = 3.98$ J

KE lost = 800 - 3.98 = **796 J** (99.5% lost — perfectly inelastic)

6. Work, Energy, and Power

Work Done

$$W = Fd\cos\theta$$

where θ is the angle between the force and the displacement. Units: J (joules). If the force is perpendicular to the displacement, no work is done (cos 90° = 0). Work done against friction is transferred to thermal energy.

Kinetic Energy and Gravitational Potential Energy

$$E_k = ½mv^2$$

$$\Delta E_p = mg\Delta h$$

Derivation of E_k = ½mv^2: From W = Fs and F = ma and $v^2 = u^2 + 2as$ (with u = 0): $s = v^2/(2a)$. So W = ma × $v^2/(2a)$ = ½mv^2.

Conservation of Energy

Energy cannot be created or destroyed, only transferred between stores. In the absence of friction/air resistance:

$$\text{Loss in GPE} = \text{Gain in KE: } mg\Delta h = ½mv^2$$

Worked Example: A roller coaster car (800 kg) descends 25 m from rest. Assuming no friction, find its speed at the bottom.

mgh = ½mv^2 (mass cancels)
v = √(2gh) = √(2 × 9.81 × 25) = √490.5 = **22.1 m s^{-1}** (about 80 km/h)

Power

$$P = W / t = \Delta E / t$$

$$P = Fv$$

Units: W (watts) = J s^{-1}. The equation P = Fv is useful for finding the driving force at a given speed, or the power needed to maintain a constant speed against resistive forces.

Efficiency

$$\text{Efficiency} = \text{useful output energy (or power)} / \text{total input energy (or power)}$$

Often expressed as a percentage. Always less than 100% in real systems due to energy dissipation (usually thermal).

Worked Example: An electric motor lifts a 50 kg mass through 3.0 m in 4.0 s. The motor draws 500 W from the supply. Calculate the efficiency.

Useful power output = mgh/t = (50 × 9.81 × 3.0) / 4.0 = 1471.5 / 4.0 = 368 W
Efficiency = 368 / 500 = 0.736 = **73.6%**

7. Density

$$\rho = m / V$$

Units: kg m^{-3}. Density is a property of the material, not the object. Typical values: water = 1000 kg m^{-3}, steel = 7800 kg m^{-3}, air = 1.2 kg m^{-3}, gold = 19300 kg m^{-3}.

8. Hooke's Law and Springs

Hooke's Law: The extension of a spring is directly proportional to the applied force, provided the limit of proportionality is not exceeded.

$$F = kx$$

- F = force (N)
- k = spring constant (N m^{-1}) — a measure of stiffness (higher k = stiffer spring)
- x = extension (or compression) from natural length (m)

Elastic Potential Energy

$$E_e = ½Fx = ½kx^2$$

This is the area under the force-extension graph (a triangle for a Hookean spring). This only applies up to the limit of proportionality.

Worked Example: A spring with k = 40 N m^{-1} is stretched by 0.15 m. Calculate (a) the force, (b) the elastic PE stored.

(a) F = kx = 40 × 0.15 = **6.0 N**

(b) $E_e = ½kx^2 = ½ \times 40 \times 0.15^2 = ½ \times 40 \times 0.0225$ = **0.45 J**

Springs in Series and Parallel

Arrangement	Combined Spring Constant	Effect
Series	$1/k_{total} = 1/k_1 + 1/k_2$	Softer overall (lower k, greater extension for same force)
Parallel	$k_{total} = k_1 + k_2$	Stiffer overall (higher k, less extension for same force)

Exam Tip: Springs in series/parallel combine the OPPOSITE way to resistors. Springs in series are like resistors in parallel (reciprocals add). Springs in parallel are like resistors in series (values add directly).

Loading and Unloading

- Below the **elastic limit**: the material returns to its original length when the force is removed. No permanent deformation. Energy is fully recoverable.
- Beyond the **elastic limit**: permanent (plastic) deformation occurs. The material does not fully return to its original length.
- For a **rubber band**, the loading and unloading curves do not follow the same path (hysteresis). The area between the curves represents energy dissipated as heat.
- For a **metal wire** below the elastic limit, loading and unloading curves overlap — all stored elastic energy is recovered.

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9. Stress, Strain, and Young's Modulus

Tensile stress (σ): Force per unit cross-sectional area. $\sigma = F / A$. Units: Pa ($N\ m^{-2}$).

Tensile strain (ε): Extension per unit original length. $\varepsilon = \Delta L / L$. No units (dimensionless ratio).

Young's modulus (E): The ratio of stress to strain within the proportional region. $E = \sigma / \varepsilon = (FL) / (A\Delta L)$. Units: Pa. This is a material property independent of the specimen dimensions.

Worked Example: A steel wire of length 2.0 m and diameter 0.50 mm supports a load of 80 N. The wire extends by 1.6 mm. Calculate (a) the stress, (b) the strain, (c) the Young modulus.

(a) $A = \pi(d/2)^2 = \pi(0.25 \times 10^{-3})^2 = 1.963 \times 10^{-7}\ m^2$
$\sigma = F/A = 80 / 1.963 \times 10^{-7} =$ **4.08×10^8 Pa = 408 MPa**

(b) $\varepsilon = \Delta L/L = 1.6 \times 10^{-3} / 2.0 =$ **8.0×10^{-4}** (dimensionless)

(c) $E = \sigma/\varepsilon = 4.08 \times 10^8 / 8.0 \times 10^{-4} =$ **5.1×10^{11} Pa = 510 GPa**
(Note: typical steel is ~200 GPa; this high value suggests the measurement has errors)

Stress-Strain Graphs

Feature	Description
Limit of proportionality	Beyond this, stress is no longer proportional to strain (Hooke's law fails)
Elastic limit	Beyond this, permanent deformation occurs on unloading
Yield point	A small increase in stress causes a large increase in strain (material flows)
Ultimate tensile stress (UTS)	The maximum stress a material can withstand

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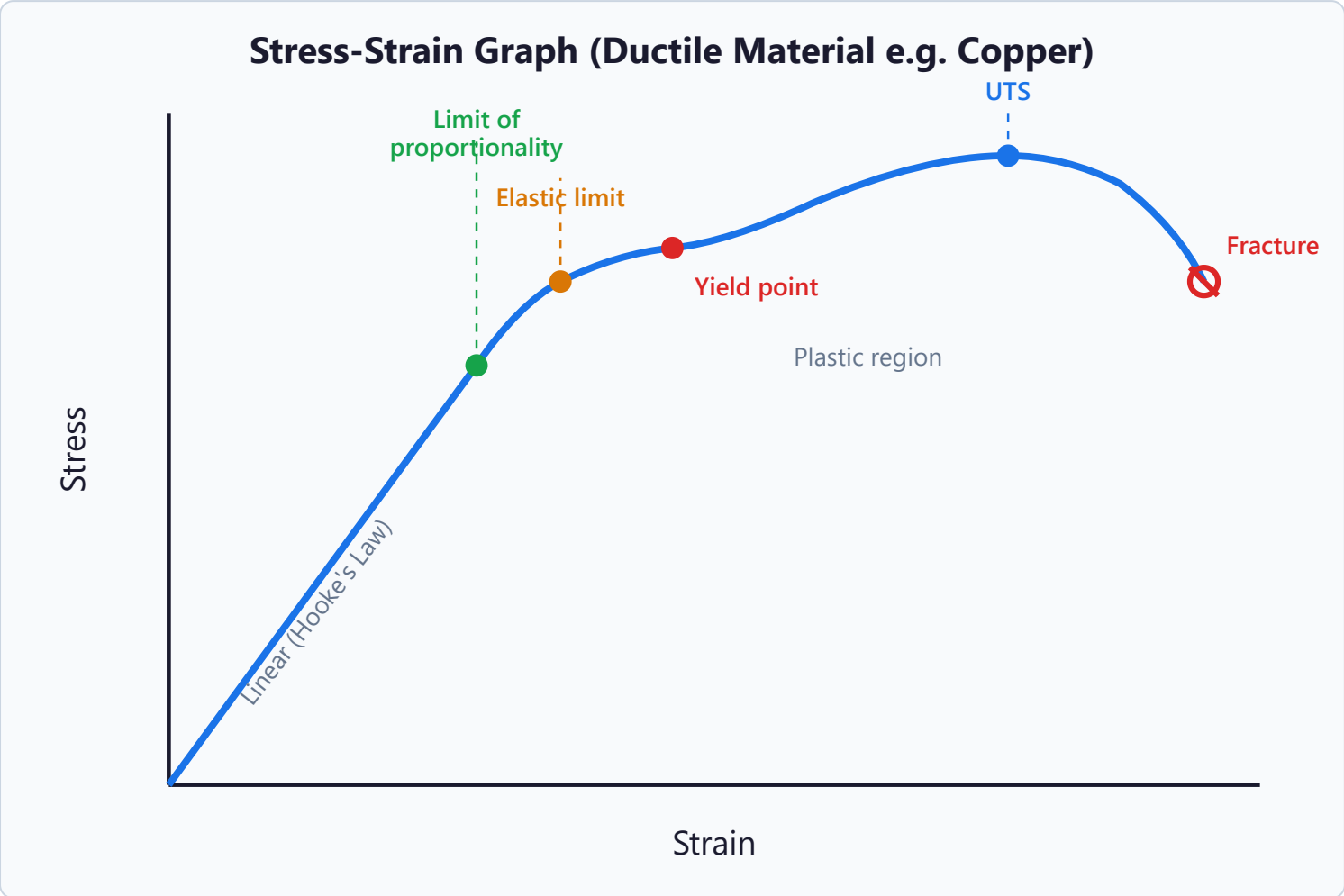
Fracture point	The stress at which the material breaks



Material Behaviour

Material Type	Characteristics	Example	Application
Ductile	Large plastic deformation before fracture. Can be drawn into wires. Clear yield point.	Copper, mild steel, aluminium	Electrical wiring, car body panels
Brittle	Little/no plastic deformation. Fractures suddenly. Straight stress-strain line until snap.	Glass, ceramics, cast iron	Windows, pottery, engine blocks

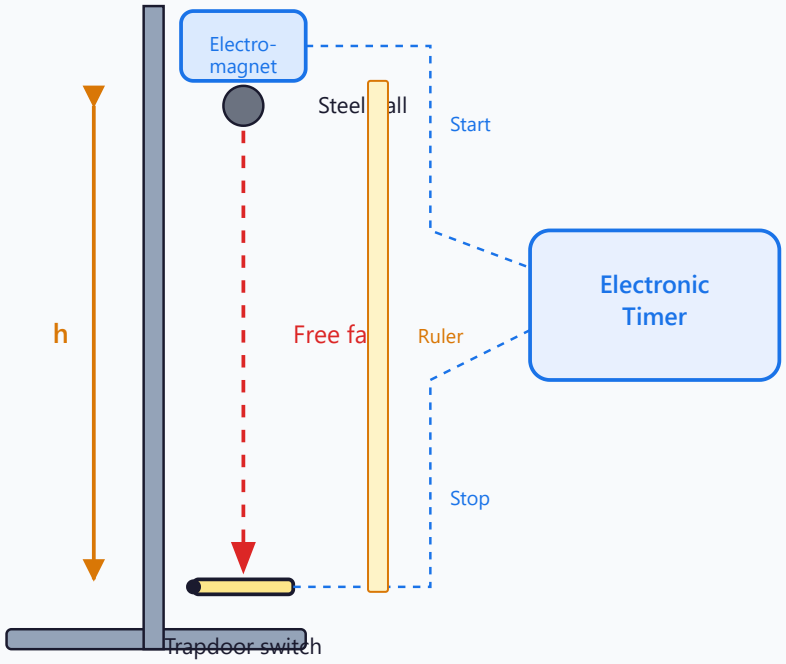
Polymeric	Very large strains possible. Behaviour depends on type.	Rubber (elastic), polythene (plastic)	Tyres, packaging

Real-world application: Aircraft fuselage panels use aluminium alloys (ductile — they deform rather than shatter on impact). Bridge cables use high-tensile steel (high UTS). Bone is a composite material — it has properties of both brittle (ceramic-like mineral) and tough (collagen fibres) materials.

Exam Tip: The area under a stress-strain graph = energy stored per unit volume (J m^{-3}). The area under a force-extension graph = energy stored or work done (J). The gradient of the linear region of a stress-strain graph gives the Young modulus. Know these three facts — they come up frequently.

⚗ Required Practical 3: Determination of g by Free-Fall

RP3: Determination of g by Free-Fall



Apparatus

- Electromagnet connected to a low-voltage supply
- Steel ball bearing

- Trapdoor switch (or light gates as alternative)
- Electronic timer (millisecond precision)
- Metre ruler or tape measure
- Retort stand, clamp, cushion/sand tray

Method

1. Set up the electromagnet at measured height h above the trapdoor. Timer starts when the electromagnet switches off (releasing ball) and stops when ball hits trapdoor.
2. Release the ball. Record time t. Repeat 3-5 times at this height for a mean t.
3. Change h in steps (e.g. 0.2 m to 1.0 m in 0.1 m increments). Repeat timing at each.
4. Measure h from bottom of ball to top of trapdoor.
5. Plot h (y-axis) against t^2 (x-axis). Gradient = g/2, so g = 2 × gradient.

Variables

Independent	Dependent	Control
Height h	Fall time t	Same ball, same release mechanism, same measuring technique

Expected Results

From $h = ½gt^2$, the graph of h vs t^2 is a straight line through the origin with gradient g/2. Expected g = 9.81 m s^{-2}.

> **Worked Example:** The gradient of h vs t^2 is measured as 4.85 m s^{-2}. Calculate g and the percentage difference from the accepted value.
>
> g = 2 × gradient = 2 × 4.85 = **9.70 m s^{-2}**
> % difference = |9.81 - 9.70| / 9.81 × 100 = **1.12%** (good accuracy)

Safety

- Cushion beneath trapdoor to catch ball

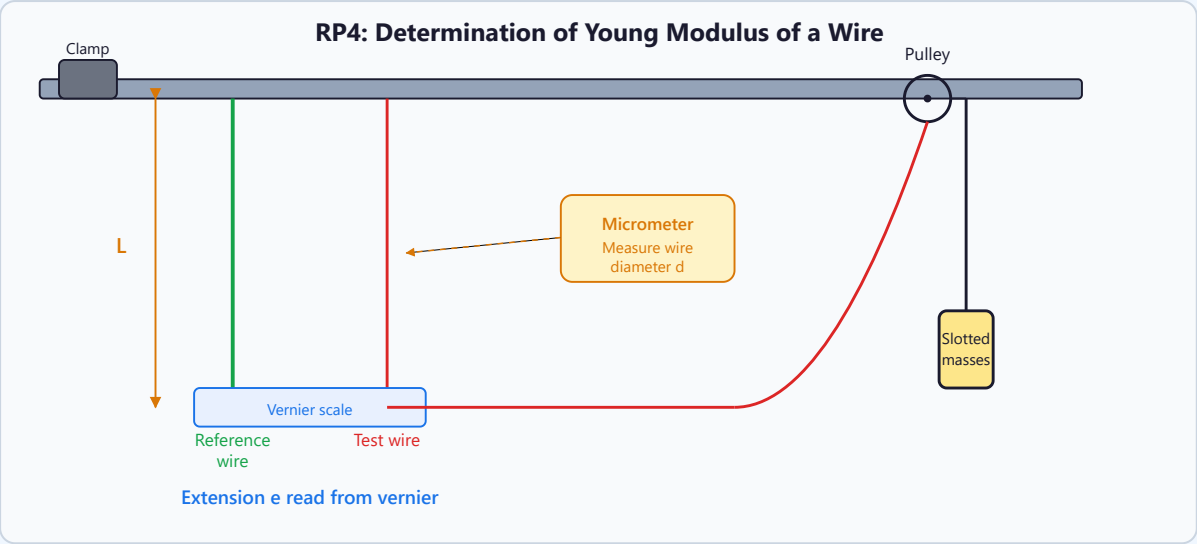
- Secure retort stand to prevent toppling
- Low voltage for electromagnet

Sources of Error and Improvements

Source of Error	Type	Improvement
Residual magnetism delays ball release	Systematic (g too low)	Use light gates or mechanical release instead
Parallax in height measurement	Random	Use set square; view at eye level
Air resistance on ball	Systematic (g appears lower)	Use small, dense ball; keep h relatively short
Small t at short heights gives large % uncertainty	Random	Use larger heights; plot graph to find g from gradient

Exam Tip: The electromagnet introduces a systematic error (residual magnetism delays release). This makes t too long and g too low. If the graph of h vs t^2 does not pass through the origin, this indicates a systematic timing error. Light gates avoid this problem entirely.

⚗ Required Practical 4: Determination of Young Modulus of a Wire



Apparatus

- Long thin copper or steel wire (at least 2 m) and reference wire of same material
- Vernier scale (to measure extension to 0.1 mm)
- Micrometer screw gauge (to measure wire diameter to 0.01 mm)
- Metre ruler/tape measure (for original length L)
- Slotted masses and hanger, pulley, G-clamp, safety goggles

Method

1. Clamp both wires to the same rigid support. Reference wire carries a small fixed mass. Test wire passes over pulley with mass hanger.
2. Measure original length L of test wire from clamp to vernier marker.
3. Measure diameter d with micrometer at several points and orientations. Calculate mean d and area $A = \pi(d/2)^2$.
4. Record initial vernier reading (with hanger only, no added masses).
5. Add masses in equal increments (e.g. 100 g). After each, read vernier. Extension e = new reading - initial.
6. Continue up to safe maximum (do not exceed elastic limit).
7. Unload in reverse to check for permanent deformation (loading and unloading should overlap).
8. Plot stress ($\sigma = F/A$) vs strain ($\varepsilon = e/L$). Gradient of linear region = Young modulus E.

Safety

- **Wear safety goggles** — wire may snap under tension

- Place cushion beneath masses; stand to the side, not below the wire
- Do not exceed elastic limit

Sources of Error and Improvements

Source of Error	Type	Improvement
Wire diameter varies along length	Random	Measure at several positions and orientations; take mean
Extension very small (fractions of mm)	Random	Use long wire (2+ m) to maximise extension; use vernier scale
Wire not initially straight (kinks)	Systematic	Pre-load with small mass before recording initial length
Thermal expansion	Systematic	Reference wire compensates for temperature changes
Parallax reading vernier	Random	Use travelling microscope for greater precision

Exam Tip: The reference wire compensates for thermal expansion and building vibrations. The biggest uncertainty is usually the diameter — since area depends on d^2, a 1% error in d causes a 2% error in A and hence E. Always state you measure diameter at multiple points and orientations.